Mooketsi Vincent Magwaza

Full-Stack Engineer · Backend, Product Systems & Platform Engineering
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PROFESSIONAL SUMMARY

Product-minded full-stack engineer who turns real operational problems into working software across interface, API, data, security, documentation, deployment, and observability. Experience includes offline-first geospatial systems, browser-based inventory and expiry tracking, distributed Rust services, local-first desktop software, PostgreSQL/PostGIS, CI/CD, and Docker-based operations. Strongest where user workflow, data integrity, and system boundaries must be designed together.

SELECTED PROJECT EXPERIENCE

Kori - Offline-First Wildlife Platform

2026

Product design, full-stack engineering, geospatial data, privacy, and operations

- Built four Next.js/TypeScript apps for low-connectivity safari use: a phone field app, landing site, review console, and API reference; sightings queue in IndexedDB and sync when connectivity returns.
- Implemented MapLibre and PMTiles offline maps, QR sharing, on-device photo metadata removal, review delays, coarse spatial cells, and track-only routing that excludes sensitive species.
- Built and privately verified a Savuti pack from OpenStreetMap tracks and Sentinel-2 imagery; satellite-first rendering, attribution, local storage, routing, and offline reload passed 16/16 automated checks.
- Developed a FastAPI/PostGIS backend with Alembic migrations, accounts, scoped API keys, uploads, reviews, trips, lifecycle jobs, Docker Compose, OpenTelemetry, Prometheus, Grafana, Tempo, backup tooling, and CI. The backend is tested but not yet connected to the apps.

Tagwise - Browser-Based Asset, Stock & Expiry Register

2026

Product strategy, React application, spreadsheet workflows, Rust services, and data integrity

- Started from a pharmacy problem: my sister needed a practical way to track batches, bulk stock, and expiry dates without buying a desktop POS subscription plus separate extensions.
- Built a responsive React/TypeScript workspace with an Excel-like register, saved views, QR/barcode scanning, printable labels, batch and expiry fields, expiring-stock alerts, multi-person stock-takes, roles, history, and stale-edit conflict handling.
- Implemented Excel/CSV import with column matching, date and duplicate validation, preview-before-write, problem reports, and formula-injection guards; the browser build works with a labelled local sample workspace and requires no dedicated desktop app.
- Built five Axum services in Rust behind nginx with PostgreSQL, Redis, migrations, concurrency/idempotency tests, OpenAPI docs, Docker Compose, GitHub Actions, and OpenTelemetry/Prometheus/Grafana/Tempo. The web app is not yet connected to these services.

Opaline - Local-First Windows Productivity Workspace

2026

Tauri 2, Rust, TypeScript, CSS, WebView2, and Windows APIs

- Built a full-screen acrylic productivity workspace combining organised Desktop panes, custom home screens, installed-app search, a taskbar mirror, local screen-time history, Pomodoro focus, Zen prompts, and a calm screen saver.
- Defined a strict safety boundary: Opaline reads Desktop files but never moves, renames, or deletes them and makes no network requests; shipped v0.2.1 as NSIS, MSI, and portable Windows builds.

University CS Docs - Guided Computer-Science Learning Platform

2026

Next.js, MDX, reusable learning components, CI, CodeQL, and contributor documentation

- Built and deployed a course-from-scratch learning platform for University of Botswana computer science, with sequenced lessons, worked examples, browser code playgrounds, cheat sheets, flashcards, quizzes, chapter assessments, resources, progress, and focus tools.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Rust, Python, SQL, Go, Java
Frontend	React, Next.js, Vite, responsive UI, accessibility, IndexedDB
Backend	Axum, FastAPI, REST APIs, authentication, background jobs
Data	PostgreSQL, PostGIS, Redis, SQLAlchemy, sqlx, Alembic
DevOps & Observability	Docker, nginx, GitHub Actions, OpenTelemetry, Prometheus, Grafana, Tempo, Trivy
Native & Geospatial	Tauri 2, Windows APIs, WebView2, MapLibre, PMTiles